



Date: 29/12/24

Lab Practical #04 :

Perform steganography and DOS attack.

Steganography :

It is the practice of hiding information within other, seemingly innocuous data, such as images, audio files, or text, so that it is not easily detectable. The goal is to hide the existence of the information itself, in contrast to encryption, which merely secures the data but makes its presence obvious. Steganography is commonly used for covert communication, digital watermarking, and protecting sensitive information from unauthorized access.

▪ **Steganography using Steghide :**

- **Steghide** is a popular, open-source steganography tool that allows you to hide files within other files, such as images (JPEG, BMP), audio files (WAV, AU), and more. It uses encryption to ensure that the hidden data is not easily extracted without the correct password. Steghide is command-line based, making it suitable for both beginners and advanced users.
- **Key Features of Steghide:**
 - **Encryption:** It supports AES encryption for hiding data, which adds a layer of security to the hidden files.
 - **File Type Support:** Steghide can embed data into various formats, including:
 - Images: JPEG, BMP
 - Audio: WAV, AU
 - **Compression:** It automatically compresses the data before embedding, saving space.
 - **Password Protection:** You can specify a password to encrypt the hidden data, making it harder for unauthorized users to extract the data.
 - **Extraction:** It allows you to extract hidden data from an image or audio file.
- **Installation :** `sudo apt install steghide`
- **Basic Usage :**
 1. **Embedding Data (hiding) :** To hide a file (e.g., `secret.txt`) inside an image file (e.g., `cover_image.jpg`), use the following command:
 - `steghide embed -cf cover_image.jpg -ef secret.txt`
 - **-cf cover_image.jpg:** Specifies the cover file (the file in which the data will be hidden).
 - **-ef secret.txt:** Specifies the file to be hidden inside the cover file.
 - You will be prompted to enter a password for encryption (this password will be required later to extract the data).

Date: 29/12/24

```
(kali@root)-[~/stegdemo]
$ ls
steg.jpg  stegmain.jpg  text.txt
(kali@root)-[~/stegdemo]
$ steghide embed -cf stegmain.jpg -ef text.txt
Enter passphrase:
Re-Enter passphrase:
embedding "text.txt" in "stegmain.jpg" ... done
```

2. **Extracting Data** :To extract the hidden data from the cover file (e.g., `cover_image.jpg`), use the following command:

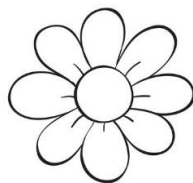
- `steghide extract -sf cover_image.jpg`
- **-sf cover_image.jpg**: Specifies the stego file (the file containing the hidden data).
- If the data was encrypted with a password during embedding, you will be prompted to enter the password.

```
(kali@root)-[~/stegdemo]
$ steghide extract -sf stegmain.jpg
Enter passphrase:
the file "text.txt" does already exist. overwrite ? (y/n) y
wrote extracted data to "text.txt".
```

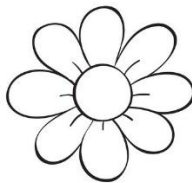
3. **Compare Data** :To compare `img1.jpg` and `img2.jpg`, then it creates a new image, `difference.png`, that shows the differences. The areas of the images that are different will be highlighted in the resulting image.

- `Compare img1.jpg img2.jpg difference.jpg`

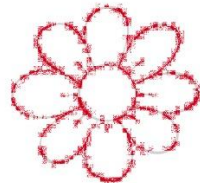
```
(kali@root)-[~/stegdemo]
$ compare stegmain.jpg steg.jpg compared.jpg
```



Stegmain.jpg



steg.jpg



compared.jpg



Date: 29/12/24

DOS Attack :

A Denial of Service (DoS) attack is a malicious attempt to disrupt the normal function of a targeted server, service or network by overwhelming it with a flood of illegitimate requests. This results in the exhaustion of resources, causing the system to slow down or become completely inaccessible to legitimate users. Unlike other forms of cyberattacks that seek to steal data, DoS attacks aim to cripple operations and degrade the availability of services.

How DoS Attacks Work:

DoS attacks exploit vulnerabilities in network protocols or system resources. Attackers may send massive volumes of traffic or specific requests that consume processing power, bandwidth, or memory. Common techniques include:

- Volume-Based Attacks: Flooding the target with traffic (e.g. ICMP flood, UDP flood).
- Protocol Attacks: Exploiting weaknesses in protocols (e.g. SYN flood, Ping of Death).
- Application-Layer Attacks: Targeting specific applications (e.g. HTTP GET/POST flood).

Distributed Denial of Service (DDoS) Attack:

A advanced form of DoS, DDoS involves multiple compromised devices (botnets) attacking a single target, making it harder to mitigate as the attack originates from different IP addresses globally.

Applications Of DoS :

DoS attacks pose significant threats to organizations, disrupting operations and causing financial losses. However, understanding and mitigating such attacks is vital in strengthening cybersecurity.

- Business Disruption
- Extortion and Ransom
- Diversionary Tactics
- Political and Activist Motives

Mitigation Techniques:

- Traffic Filtering: Implementing firewalls, intrusion prevention systems (IPS) and anti-DDoS solutions to filter out malicious traffic.
- Rate Limiting: Restricting the number of requests a server can handle from a single IP address.
- Redundancy and Load Balancing: Distributing traffic across multiple servers to prevent resource exhaustion.
- Monitoring and Anomaly Detection: Continuously monitoring network traffic to detect and respond to unusual patterns promptly.